

Unit 3.4 The Chain rule

Note Title

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Used for differentiating composite functions:

eg $\sqrt{x^2+1}$, $\sin(4x)$, e^{x^3+2x} , $(2x+e^{5x})^6$

The rule:

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$$

Special case: $\frac{d}{dx} (f(x))^n = n(f(x))^{n-1} \cdot f'(x)$

$$\frac{d}{dx} \sqrt{x^2+1} = \frac{1}{2} (x^2+1)^{-1/2} \cdot 2x$$

$$\frac{d}{dx} \sin(4x) = \cos(4x) \cdot 4$$

$$\frac{d}{dx} e^{x^3+2x} = e^{x^3+2x} \cdot (3x^2+2)$$

$$\frac{d}{dx} (2x+e^{5x})^6 = 6(2x+e^{5x})^5 \cdot (2+e^{5x} \cdot 5)$$

Fruit salad

$$\frac{1}{2} (2x)^{-1/2}$$

Families of functions:

① $\frac{d}{dx} 3\cos x = -3\sin x$

$$\frac{d}{dx} \cos 3x = -\sin 3x \cdot 3$$

$$\frac{d}{dx} \cos^3 x = 3(\cos x)^2 \cdot (-\sin x)$$

$$\frac{d}{dx} \cos(x^3) = -\sin(x^3) \cdot 3x^2$$

② $\frac{d}{dx} e^{\sin x} = e^{\sin x} \cdot \cos x$

$$\frac{d}{dx} \sin(e^x) = \cos(e^x) \cdot e^x$$

$$\frac{d}{dx} e^{e^x} = e^{e^x} \cdot e^x$$

③

$$\frac{d}{dt} \tan \sqrt{t} = \sec^2 \sqrt{t} \times \frac{1}{2} t^{-1/2}$$

$$\frac{d}{dt} \sqrt{\tan t} = \frac{1}{2} (\tan t)^{-1/2} \cdot \sec^2 t.$$

$$\frac{d}{dt} \frac{1}{\sqrt{\tan t}} = -\frac{1}{2} (\tan t)^{-3/2} \cdot \sec^2 t.$$

$$\approx (\tan t)^{-1/2}$$

A new derivative

$$\boxed{\frac{d}{dx} a^x = \ln a \cdot a^x}$$

eg $\frac{d}{dx} (2^x) = 2^x \cdot \ln 2$
0,69.

Why?

$$\frac{d}{dx} a^x = \frac{d}{dx} (e^{\ln a})^x = \frac{d}{dx} e^{\ln a \cdot x} = e^{\ln a \cdot x} \cdot \ln a$$

$$= a^x \cdot \ln a.$$

Examples:

$$\frac{d}{du} 4^u = 4^u \cdot \ln 4$$

$$\frac{1}{4^u} \rightarrow \frac{-4^u \cdot \ln 4}{(4^u)^2}$$

$$\frac{d}{du} 4^{-u} = (4^{-u}) \cdot \ln 4 \cdot (-1)$$

$$\frac{d}{du} \operatorname{cosec}(4^u) = -\operatorname{cosec}(4^u) \cdot \cot(4^u) \cdot 4^u \cdot \ln 4$$

$$\frac{d}{du} 4^{\operatorname{cosec} u} = 4^{\operatorname{cosec} u} \cdot \ln 4 \cdot (-\operatorname{cosec} u \cdot \cot u).$$

Mixed examples (sum, product & quotient rules mixed)

$$\frac{d}{dx} \left(3e^{x^2} + 6 \cos \frac{1}{x} \right) = 3 \cdot e^{x^2} \cdot 2x - 6 \sin \left(\frac{1}{x} \right) \cdot \left(-\frac{1}{x^2} \right)$$

$$\frac{d}{dx} 3^{\sin 2x} \cdot \cot(e^{-2x}) = 3^{\sin 2x} \cdot \ln 3 \cdot \cos 2x \cdot 2 \cdot \cot(e^{-2x}) + 3^{\sin 2x} \cdot (-\operatorname{cosec}^2(e^{-2x}) \cdot e^{-2x} \cdot (-2))$$

$$\frac{d}{dx} \frac{e^{2x}}{3^x - \tan \sqrt{x}} = \frac{e^{2x} \cdot 2(3^x - \tan \sqrt{x}) - e^{2x} (3^x \cdot \ln 3 - \sec^2 \sqrt{x} \cdot \frac{1}{2\sqrt{x}})}{(3^x - \tan \sqrt{x})^2}$$

$$\frac{d}{dx} \left(\frac{4}{e^{2x} + 5^{3x}} \right) =$$

- Homework sheet
- Technique Mastering Exercise I p 42.